جامعة المنوفية/

كلية الحاسبات والمعلومات

Student Tutoring Matchmaking System (STMS)

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Project Introduction

Project Identification

Project name: *Student Tutoring Matchmaking System (STMS)*

Course: System Analysis & Design

Definition:-

A web and mobile platform that matches students with suitable tutors based on subject, level, availability and ratings, and manages booking, sessions and feedback.

Objectives:-

- **Create student and tutor profiles with subjects, levels and availability.**
- **Match students with suitable tutors automatically (subject, level, availability, rating).**
- **Allow one-tap session booking with a booking calendar and automatic reminders.**
- **Support in-app chat and video-call sessions between students and tutors.**
- **Collect ratings and reviews after each session to maintain tutor quality.**
- **Give administrators a dashboard to approve tutors, handle complaints and view reports.**

Problems Solved:-

- **Students struggle to find a suitable, trusted tutor (word of mouth, social-media groups, notice boards).**
- **Arranging session times by chat is slow, and schedule mismatches and no-shows are common.**
- **No feedback mechanism, and tutors have no steady way to reach students.**

System Request–Student Tutoring Matchmaking System Project
Project Sponsor: E-Learning Platform Provider (Educational Technology Company)
Business Need: Students find tutors through word of mouth, social-media groups or notice boards, so tutor quality and availability cannot be verified and arranging times by chat is slow. Tutors have no steady way to reach students, and there is no feedback mechanism. This project aims to build a platform that matches students with suitable tutors, simplifies booking and session management, and gives administrators control over tutor approval and quality. A web and mobile platform that matches students with suitable tutors based on subject, level, availability and ratings, and manages booking, sessions and feedback.
Business Requirements: ➢ Smart Matching: Matches each student with suitable tutors based on subject, level, availability and rating. ➢ Booking and Reminders: Lets students book sessions in one tap through a booking calendar and sends automatic reminders to reduce no-shows. ➢ Tutor Verification and Approval: Verifies tutor identity and requires administrator approval before a tutor profile becomes visible.

Business Value:

- **Faster Tutor Discovery:** Automated matching saves students the time spent searching.
- **Fewer No-Shows:** One-tap booking and automatic reminders reduce missed sessions and schedule mismatches.
- **Better Academic Results:** Access to suitable tutors leads to higher grades, fewer failed courses and better student satisfaction.
- **Income for Peer Tutors:** Gives tutors a steady way to reach students and earn extra income, with an optional small commission per session to sustain the system.
- **Flexible Online Access:** 24/7 access with chat and video-call sessions, no physical rooms needed.
- **Trusted Quality:** Identity verification, admin approval and ratings help students choose reliable tutors.
- **Less Manual Administration:** Replaces notice boards, group chats and spreadsheets with automated matching, booking records and reports.
- **Cost-Effective:** First-year cost of $5,700 – $14,400 vs. annual benefits of $12,000+ (see Feasibility Study).
- **Safe Learning Environment:** A code of conduct and report/block tools protect both students and tutors.

Special Issues or Constraints:

- **Cross-Platform Access & Performance:** Must run as a web app and Android/iOS apps and handle peak demand (e.g., before exams) without slowing down.
- **System Availability & Backup:** Available 24/7 on cloud hosting with automatic backup.
- **Data Privacy & Legal Compliance:** Encryption, role-based access, parental consent for minors and a secure payment gateway.
- **Conduct & Tutor Verification:** Tutor identity verification and a code of conduct with report/block tools.
- **Timeline & User Adoption:** Delivery within an estimated 3–5 months, with short guides and training for students, tutors and administrators.

Summary:-

Element	Details
Project Sponsor	E-Learning Platform Provider (Educational Technology Company)
Business Need	Students find tutors through word of mouth, social-media groups or notice boards, so tutor quality and availability cannot be verified, arranging session times by chat is slow, and there is no feedback mechanism after sessions.
Business Requirements	Student and tutor profiles, smart matching, session booking with a calendar, automatic reminders, in-app chat and video calls, ratings and reviews, tutor identity verification and approval, and an admin dashboard for complaints and reports.
Business Value	Faster tutor discovery, fewer no-shows, higher grades and fewer failed courses, extra income for peer tutors, less manual administration, and a cost-effective solution (first-year cost of $5,700 – $14,400 against annual benefits of $12,000+).
Special Issues	Limited budget and a 3–5 month timeline, protection of personal data (some students may be minors, so parental consent is needed), secure payments, tutor verification and a code of conduct, and 24/7 availability on web and mobile.